Double and Single Slit Interference Patterns at Low and High Light Intensity

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Abstract

Double and Single-slit interference patterns were successfully obtained for a high-power laser source. The interference pattern was compared to a theoretical model and was in sound agreement. By measuring the physical properties of the slits and optical channel the wavelength of the laser was determined to be $\lambda = 632.15 \pm 3.56nm$. The optimal bias voltage across the photomultiplier was determined to be $V_{bias} = 0.548 \pm 0.01V$. Due to experimental complications no good data was obtained when the optical channel was switched to bulb-operation mode.

1 Introduction

In 1801, Thomas Young began experimenting with a class of experiments known as "double path" experiments, [1]. Double path experiments involve the splitting of a wave into two parts. When Young initially performed the experiment with two very small slits, a coherent light source and a screen, a strange pattern was observed - an interference pattern. The light did not go through one slit or the other, instead it went through both at the same time. The amplitudes of the light waves constructively or destructively combine amplitudes depending on their phase. This phenomenon displays the wave behaviour of light. The experiment was later extended by Davisson et al. [2] to show that this behaviour also occurs for electrons when the slit is many orders of magnitude smaller than that of Young's original experiment. The aim of this experiment is to demonstrate the wave-like of a single photon in the presence of a double-slit.

2 Theory

Young's original experiment showed that bright and dark spots could be observed on the screen at the positions:

$$n\lambda = d\sin(\theta) \tag{1}$$

$$\left(m + \frac{1}{2}\right)\lambda = d\sin(\theta) \tag{2}$$

where n and m are integers and represent the bright and dark spots respectively, d represents the distance between the slits and θ is the angle measured from the central maxima. These expressions are represented graphically in figure 1.

From these equations and some trigonometry, [3], the expression for the intensity of the pattern as a function of θ is

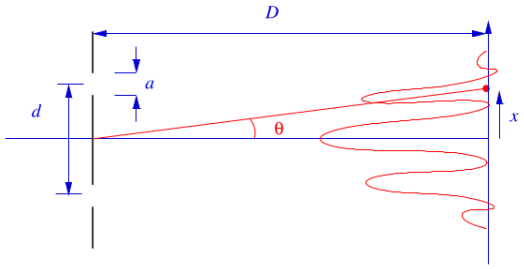


Figure 1: Double slit setup. It can be seen that to calculate α and β, and thus the intensity, the only quantities that need to be measured are d, x and λ. Figure from the lab script, [1].

derived as:

$$I(\theta) = 4I_0\cos^2\left(\frac{2\pi d}{\lambda}\sin(\theta)\right) \tag{3}$$

Later in the 19th century, Fresnel observed a similar interference pattern when line was guided through a single-slit. The positions of these minima is: [4]:

$$a\sin(\theta) = m\lambda \tag{4}$$

With this equation and more trigonometry arguments, an expression for the single-slit intensity as a function of position is derived as, [3]:

$$I(\theta) = I_0\left(\frac{\sin(\alpha)}{\alpha}\right)^2; \alpha = \frac{\pi a\sin(\theta)}{\lambda} \tag{5}$$

In the lab script these two expressions are combined as a product to produce a single equation which governs the behaviour of both the double and single-slit interference

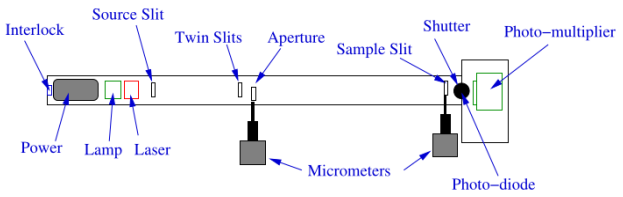


Figure 2: The optical channel provided by TechSpin. Figure from the lab script [1].

patterns, [1]:

$$I = I_0 \left(\frac{\sin(\alpha)}{\alpha} \cos(\beta) \right)^2 \tag{6}$$

where $\beta = \frac{\pi d \sin(\theta)}{\lambda}$. In this setup the angles considered will be $\sim 1^\circ$ so it is appropriate to use the small angle approximation, $\sin(\theta) \approx \theta$, which simplifies α and β to be:

$$\alpha = \frac{\pi a x}{\lambda D}; \beta = \frac{\pi d x}{\lambda D} \tag{7}$$

3 The Experiment

3.1 Experimental Apparatus

The experimental apparatus provided by TechSpin includes: an optical channel (depicted in figure 2), a laser, a bulb, tweezers, a white card, a discriminator, two single slits, two double slits and an oscilloscope. An optical microscope is provided in the laboratory for measuring the slit separations. The optical channel consists of a lamp and laser at a fixed optical path length from the photodiode. The slits are made of a ferromagnetic material so they can be attached to the channel securely. The laser has wavelength of $\lambda = 670 \pm 20nm$ when the power supplied to is less than $1.0mW$, [5]. The total optical path length from the source slit to the photo-diode is $1m$, specified without uncertainty by TechSpin, [5]. When set to laser mode, the light passes through the source slit into the fixed twin slits at which point the interference pattern begins. The aperture slit is free to move so that the slits can be blocked at will. After the light passes through one or both of the slits it goes through the sample slit, which is also free to move, and then is incident on the photo-diode - at which point the intensity is measured. When the optical channel is set to bulb mode the photo-multiplier must be turned on to get a non-zero reading at the photo-diode. The photo-multiplier amplifies the signal at the photo-diode by 100. The overall detector efficiency is 10%, so only one in ten photons incident on the photo-diode are counted.

When handling the slits a pair of tweezers must be used to avoid contamination from fingerprints/oil from skin, which will affect the path taken by the light.

3.2 Experimental Procedure

3.2.1 Calibration

Before taking any measurements the slits in the optical channel must be properly aligned so that 1. the light reaches the sample slit and 2. a large portion of the amplitude of the laser reaches the photo-diode. The steps taken to properly align the laser (after some trial and error) were as follows:

1. Attach a multimeter across the photo-diode,
2. Open the lid,
3. Turn on the laser,
4. Place the slits into the correct slots with tweezers,
5. Place a white card in front of the source slit and nudge the slit until the laser looks like it is at it's brightest on the card,
6. Place the white card in front of the double slit and nudge the slit until the interference pattern looks clearest,
7. By adjusting the micrometer attached to the aperture slit, record positions where *only* slit 1, *only* slit 2 and *both* slits are incident on the card,
8. Remove the white card and move the micrometer at the sample slit until a maximum intensity is found - record this position and ensure it is $\approx 2.0V$, [1].

The calibration procedure is identical for the bulb portion of the experiment, however, provided the setup is not disturbed in some way (i.e. bumping the table or removing the lid violently) the calibration need only be performed once.

3.2.2 Laser Measurements

The procedure for measuring the double-slit and single-slit interference patterns is almost identical. An increment for measurement must be chosen, Δx, so that the interference pattern can be plotted digitally. The value chosen for the double slit experiment was $\Delta x = 0.25mm$ and for the single slit was $\Delta x = 0.20mm$: ideally these values should be the same but there were time related issues which will be discussed in section 6. When the micrometer is set to $x = 0$ the intensity increases linearly because the body of the sample slit is not fully covering the photo-diode, thus great care must be taken to start at a point where the only light reaching the photodiode comes through the sample slit - in this case this is $0.5mm$. Again, due to time issues the full range was not explored in the double slit measurement - these start at $x = 4.45mm$. The same effect happens when the body of the sample slit stops covering the other end of the photodiode, which is somewhere around $x \approx 9.3mm$, depicted in figure 4. At each increment the displacement, x, is recorded from the micrometer and the voltage at the photodiode is recorded at the multimeter. When a maxima

is encountered an additional data point is added regardless of what value x takes at this point. This procedure is repeated for the single-slit (both slits are measured separately) and double-slit interference patterns.

3.2.3 Bulb Measurements

As with the laser measurements, the optical channel must be properly aligned before measurements can be taken. After this is ensured, the optical channel can be switched from laser mode to bulb mode at the power source. Then the laser must be unscrewed so that the light from the bulb is not obscured by it. With the lid secured on the channel the photomultiplier can be switched on, which amplifies the signal by a factor of 100 so it can easily be observed using a standard multimeter. Now, before measurements can be made, the discriminator must be connected to the photodiode and calibrated. At this point, the signal will also be passed through an oscilloscope to verify that background counts are being accurately distinguished from photons incident on the photodiode. To calibrate the discriminator, the bulb is set to power mode 7 (high power) and the discriminator gate is set to 10 seconds. Measurements were recorded for a period of 50 seconds (5 total measurements) with the shutter both up and down (the shutter closed should be purely background counts). The five sets of measurements are averaged and then the ratio of counts to noise is calculated. The nature of these background counts will be discussed in section 6. The optimal value for the bias voltage was thus determined by incrementally varying the bias voltage first by $100mV$, then $10mV$ and then finally $1mV$ until this ratio was maximized. With the bias voltage set to the optimal value, the double-slit and single slit measurement procedure is now identical to that of the laser measurements (3.2.2) - except the frequency of counts incident on the photodiode is now measured by the discriminator instead of voltage across it. The entire range of the micrometer was explored in increments of Δx, noting down the frequency at these values. The positions of all maxima are also recorded. These measurements are first taken with the bulb at a power of 7 to demonstrate that the interference pattern is still present at low intensities - then the bulb power is lowered to 0 for measurements in "single photon operation mode".

3.3 Single Photon operation

To confirm that only one single photon is in the optical channel at any time, the frequency reading on the discriminator was used. When the value is of the order of $\sim 1Hz$, the number of photons in the system is ≤ 10 (10% detection efficiency). Given the fact that the optimal signal to noise ratio was determined to be ≈ 10, a reading of $\sim 1Hz$ should correspond to ≈ 1 photon in the system at this frequency reading.

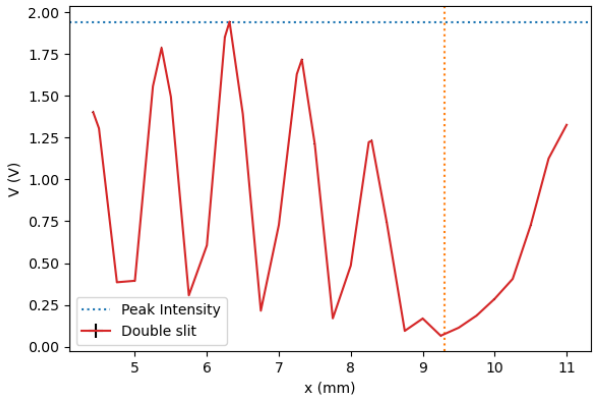


Figure 3: Interference pattern in double-slit operation mode. The interference pattern is observed to represent the cosine term in the intensity function, 6. Similar to the single slit graphs, the intensity has a linear growth at $x \approx 9.3mm$. Error bars are included in the plot but are too small to see.

4 Results

All raw data will be submitted in spreadsheet format and all python scripts used in calculations are also included. Five of the six key results in this experiment were obtained - 3 of which are purely graphical results.

4.1 Double and Single Slit

The first of these results is the confirmation of an interference pattern when the high-power laser is directed through the double-slit and single-slit configurations of the optical channel. The results are shown in figures 3 and 4. It should be noted that the double slit result was rushed and thus the curve is not smooth - but still identifiable as a sinusoidal waveform. The maximum intensity of the double slit setup does not equal the sum of the maximum intensities of the two single slit results (around $\sim 0.5V$ short), this is likely due to the aperture slit blocking more light than intended due to improper alignment (not much thought was put into this during the lab session so this explanation could be incorrect). It can be seen that on all 3 plots the intensity begins to follow a linear growth at $x \approx 9.3mm$, this is because the sample slit begins to leave portions of the photodiode exposed to light that does not enter through the sample slit. Therefore, measurements after $x \approx 9.3mm$ can be ignored. This also occurs when the sample slit is at the other extreme position, when $x \approx 0.5mm$. Measurements were conducted for $x \geq 0.5mm$ to avoid this effect.

4.2 Theoretical Comparison

The second result was simply to compare the first result to the theoretical prediction by measuring the physical dimensions of

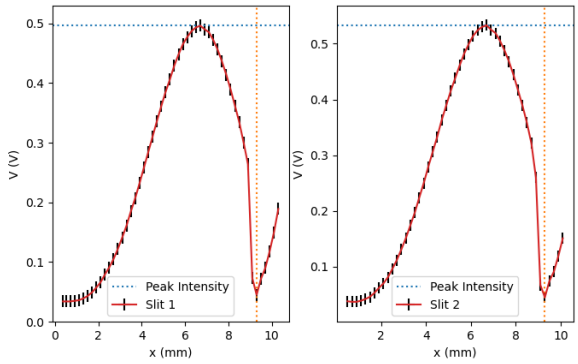


Figure 4: Interference pattern in single-slit operation mode. It can be observed that an interference pattern is present and represents the sine term in the intensity function, 6. The maximum intensity is slightly different for each of the slits, and when the two values are summed the result is less than the peak intensity in the double slit graph 3. The orange dotted line represents the point where the sample slit is no longer fully covering the photodiode.

the slits and optical channel. Equation 6 was then modelled in in python in figure 5. The location of the measured peaks are in good agreement with the theoretical peaks (within $\sim 3\sigma$).

4.3 Measuring the Wavelength of the Laser

To measure the wavelength of the laser, Young's equation for the double slit experiment was used, 1. Measurements were made for the 0th maxima and gave a wavelength of $\lambda = 632.15 \pm 3.56nm$. This result differs quite considerably from the TechSpin result ($\lambda = 670 \pm 20nm$), which is likely because the optical path length between the double slit and the photodiode, D, were not measured in the lab (see section 6).

4.4 Optimize the Bias Voltage Across the Photo-Multiplier

The optimal bias voltage was first determined to be $0.530 \pm 0.01V$ but after consulting a TA, the method was changed and the value was finalized to be $V = 0.548 \pm 0.001V$. The latter method corresponds to a signal-to-noise ratio ~ 3 times higher than the former.

4.5 Bulb-operation mode Double and Single-slit

The results for this section are purely graphical and displayed in figure 6. Two attempts were made to observe an interference pattern with the bulb on high-power mode and both attempts were unsuccessful. The first attempt was incorrectly done without the attenuation mode on the discriminator,

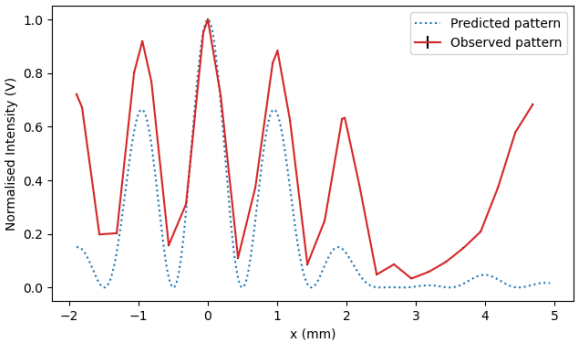


Figure 5: Observed double slit laser fringe pattern plotted against the theoretical prediction. It can be seen that maxima are in phase with the theoretical prediction. Discrepancy in intensity values will be discussed in section 6. Error bars are included in the plot but are too small to see.

which explains the significantly higher peaks - caused by electrical noise in the discriminator. The second attempt also did not show any signs of an interference pattern, likely due to the misalignment of the laser at some point between the optimization of the bias voltage and these measurements. Due to time limitations this problem was never resolved (even after consulting a TA) and the single slit interference pattern could not be tested.

5 Error Analysis

5.1 Double and Single Slit Measurements

The measurements for this result rely only on the micrometer at the sample slit, which has an instrumental uncertainty of $\pm 0.005mm$, and the multimeter, which has an instrumental error of $\pm 0.01V$. These errors were included as error bars in figures 3 and 4. Both error bars are so small that they are not visible in the double-slit plot. The vertical error bars can clearly be observed in the single slit plots because the maximum peak is less than half of that in the double-slit plot.

5.2 Theoretical Comparison

This is result is purely a comparison of shape and so the difference in maxima position are taken as a reference point. The graphs are aligned such that the 0th order maxima occur at the same point on the x axis. Both intensities are normalised so that the 0th order maxima occurs at $1V$, this is done because *only* the shape is being considered. The error bars are the corresponding x and V errors from the original double slit plot, figure 1. The only errors present in the theoretical prediction are the resolution of the x array used to calculate the corresponding intensity values in python,

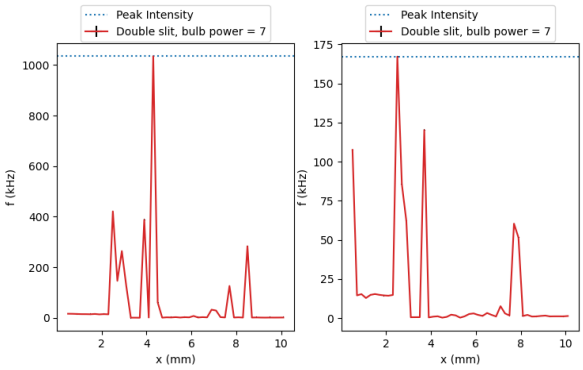


Figure 6: Plot of Frequency against displacement in bulb-operation mode. No interference pattern can be observed using an increment of $\Delta x = 20mm$. The plots represent two separate attempts at measuring the interference pattern, the data on the left was recorded when the attenuation setting was accidentally turned off on the discriminator. The right plot shows the measurements with the discriminator setting on. Error bars are included in both plots but are too small to see.

which is $\pm 0.01mm$.

5.3 The Wavelength of the Laser

The wavelength of the laser was calculated using equation 1 and the small angle approximation, $\sin(\theta) \approx \theta$. The errors are propagated using the standard calculus error propagation formula:

$$\delta\lambda = \sqrt{(\delta x)^2(\frac{\partial\lambda}{\partial x}^2) + (\delta d)^2(\frac{\partial\lambda}{\partial d}^2) + (\delta D)^2(\frac{\partial\lambda}{\partial D}^2)} \quad (8)$$

D was not measured physically but taken from the TechSpin manual as $D = 40cm$ with no error, so the δD term cancels out of the equation. After plugging in the derivates and numerical values of $\delta x = \pm 0.005mm$ and $\delta d = \pm 0.0005mm$ the error on lambda comes out to be:

$$\Delta\lambda = \pm 3.56nm \quad (9)$$

This value is $\sim \frac{1}{4}$ of the error provided by TechSpin. Measurement of the optical path length, D, in the laboratory would likely bring these results into alignment.

5.4 Bias Voltage

The Bias Voltage was optimized by calculating the ratio of counts when the shutter is up vs. when the shutter is down. The number of counts (actually reported as frequency by the discriminator) has an error $\sim \frac{1}{\sqrt{N}}$. Reading the frequency values off the screen also has an uncertainty of $\pm 0.000001Hz$. The finalised error on the voltage comes from the finite precision of the multimeter, which can clearly be seen in the method described in section 3.2.3. Thus, the error on the optimized bias voltage is $\Delta V_{bias} = \pm 0.001V$.

5.5 Bulb operation Double-Slit Measurements

The results for this section are, similar to the laser operated double-slit measurements, purely graphical and thus presented in the form of error bars. The error on the frequency measurements comes from the standard calculus error propagation formula, equation 8. It is worth mentioning that there is a contribution from the inability of computers to keep track of time consistently, the value chosen is $\Delta t = 10ms$, from [6]. The result is the error on frequency as a function of the frequency. The error bars for each individual point is calculated using python and included in the plot - figure 6.

6 Discussion

6.1 Optical Channel Alignment Concerns

Throughout the four lab sessions committed to this experiment the laser was aligned only two times. This eventually led to issues in the final lab session where the plan was to measure the double and single slit interference patterns and, time permitting, measure the interference pattern for the double slit when the optical channel was in "single photo operation mode". When attempting to measure the double-slit interference pattern for the bulb on high power mode only background noise was obtained over the entire measurable range. After consultation with a TA it was concluded that the issue was likely the fact that the attenuation button on the discriminator was not activated. After spending time troubleshooting this issue it was decided to attempt the measurements again without re-aligning the laser. This was a mistake because the data was again random noise. At this point there was no time to re-align the laser and take measurements so the final lab session concluded. If the experiment were to be repeated, the laser should be aligned at the start of *every* lab session to prevent issues like this occuring in the future.

6.2 Time Concern

The experiment was performed under significant time pressure which resulted in poor data collection for most of the presented results. A major time loss stemmed from a misunderstanding on how to align the laser properly, which came into realization half way through lab session two and was only resolved at the end of that lab session. This resulted in ~ 1.5 lab sessions worth of time being lost. The measurements for double-slit interference with the high-power laser were rushed to partially accommodate for this. Later in the experiment a fundamentally flawed attempt to determine the optimal bias voltage by using a minimal standard deviation approach was attempted. This led to a further ~ 0.75 lab session time penalty spent troubleshooting the setup. These

time penalties were ultimately caused by a combination of skim-reading the lab script and the physical lab-script being an outdated version. In future greater care will be taken when reading the lab script and online versions of the lab script will be used where available.

6.3 Measurement of Optical Path Length

The optical path length between the double slit and the photodiode was not measured during the lab sessions. This is likely the root cause of the discrepancy between TechSpin's value of λ for the laser and the one calculated in the lab using Young's equation, 1. TechSpin quote a value of $40cm$ with no error and this was used in all relevant calculations. Physically, this result being exact is very unlikely and - upon discussion with my peers who *did* measure this value - incorrect. If the experiment were to be repeated, the optical path length would be measured with a metre stick (or another similar instrument available in the laboratory).

6.4 Nature of the Background Counts

When measuring the frequency of counts incident on the photodiode in bulb-operation mode, background counts are observed. The optical channel is completely isolated from any visible light coming from the lab so this background must come from somewhere else. Background counts from cosmic rays will be minimal due to the fact that the lab is a windowless room underneath several stories of concrete. More probably, the background counts come from the discriminator itself. It can be observed that in the dataset (on the left of figure 6) recorded with the attenuation setting off, the peaks are much higher than when it is on. This suggests much less counts overall during the 10s gate periods. The attenuation setting runs the signal from the photodiode through a series of resistors which eliminates electrical noise from the wire, oscilloscope and discriminator. Another possible source of this background noise is from background beta radiation in the lab. If the beta radiation was incident on the photodiode then this could knock an electron off the positively charged plate in the photomultiplier and cause the chain reaction in the same way that a photon incident on the photomultiplier would. Considering the optical channel is sealed inside a metal container, which would block beta radiation from entering, this explanation is unlikely. Thus the background is probably electrical noise. The approach taken in this experiment to minimize this noise is to optimize the bias voltage across the photomultiplier and use the attenuation mode on the discriminator. In the future, to reduce this noise the bias voltage could be more finely optimized by using a more precise voltmeter.

7 Conclusion

To conclude, a laser was directed through an optical channel which encased a double or single slit. The aim was to measure the voltage across the photodiode at various positions. The results were plotted and agree well with theoretical predictions, which were calculated by measuring the physical dimensions of the slits and the optical path length of the laser. Then, the optical channel light source was changed to a bulb which was capable of low and high power operation. The photomultiplier in the optical channel was switched on and the voltage across it (called the bias voltage) was tuned to maximize the signal to noise ratio. An attempt was made to measure the double slit interference pattern with the bulb on high power mode but various issues were encountered which led to no successful data being collected. Lab time concluded before data collection in "single photon operation mode" could be attempted.

References

[1] School of Physics Astronomy. "Two Slit at High and Low Light Levels". *Junior Honours Lab Scripts*. (2024)

[2] C. Davisson and L.H. Germer. "The Scattering of Electrons by a Single Crystal of Nickel." *Nature*. (1927)

[3] Jearl Walker. *Haliday and Resnick's Principles of Physics* $11^{th} edition$. (2020)

[4] Wikipedia. *Diffraction*. URL: `https://en.wikipedia.org/wiki/Diffraction#Single-slit_diffraction`.

[5] TechSpin. *Two-Slit Interference, One Photon at a Time*. URL: `https://www.teachspin.com/two-slit`.

[6] David L. Mills. "Precision synchronization of computer network clocks". *SIGCOMM Comput. Commun. Rev.* (1994)